

THARINDU DAMESHA RAJAPAKSHA

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Professional Summary

PhD Candidate in Physics specializing in semiconductor devices and photonic nanostructures with over 4 years of hands-on cleanroom and nanofabrication experience with strong expertise in graphene-based devices, diffractive optics, and optoelectronic characterization.

Education

PhD in Physics, Virginia Polytechnic Institute and State University, Blacksburg, VA | 2021–Present

Master's in Physics, Virginia Polytechnic Institute and State University, Blacksburg, VA | 2021–2024

B.Sc. (Honors) in Physics, University of Peradeniya, Sri Lanka | 2016–2020

Selected Publications

1. Jahan, N., Rajapaksha, T. D., et al Extended Long-Wavelength Photodetection by Integrating Graphene with Different Bandgap Semiconductors (Submitted to Advanced Electronic Materials)
2. Ellepola, K., Rajapaksha, T. D., et al. (2026). Monte Carlo Ray-Tracing Simulations for Diffractive Optics. Optics Express 34.3 (2026): 4465-4480..
3. Howe, L., Rajapaksha, T. D., et al. (2024). High-Efficiency Multilevel Phase Lenses with Nanostructures on Polyimide Membranes. Advanced Optical Materials, 12(25), 2400847.
4. Rajapaksha, R. M. T. D., et al. (2021). Enhancement of gas sensitivity of ferric oxide thin films by adding activated carbon nanoparticles. Icon. Res. Eng. J., 4, 107-13.
5. Rajapaksha, R. M. T. D., et al. (2021). Improvement of gas sensitivity of ferric oxide thin films by adding Mn nanoparticles. Bull Mater Sci 44, 182.

PATENT

Mold-Based Cast Diffractive Optics (US20260102982A1)

Experience

Graduate Research Assistant, Virginia Tech | 2021–Present

- Fabricated hybrid graphene–Si and graphene–GaAs phototransistors for mid-IR detection.
- Designed, fabricated and characterized diffractive optical devices (Fresnel zone plates, photon sieves, metalenses)
- Collaborated with NASA Langley Research Center on photonic device development (Project SPECIES).

Skills

Semiconductor Fabrication & Processing

PVD, ALD, RIE, Photolithography, Direct writing, E-beam lithography, Wet Etching, Device Dicing, Oxidation, Device fabrication & process integration

Materials & Device Characterization

SEM, AFM, FTIR, Pump-Probe Spectroscopy, Lock-in Amplifiers, Keithley SMUs, Optical characterization and spectroscopy

Simulation & Data Analysis

Ansys Lumerical, COMSOL, Meep, Python, OriginLab, KLayout, Blender 3D, MATLAB, JMP, Data analysis, curve fitting, parameter extraction, Process optimization and experimental iteration

Projects

Multilevel Fresnel Zone Plates (FZPs) on Membranes

- Fabricated multilevel phase Fresnel zone plates on flexible polyimide membranes.
- Developed thickness-control strategies for accurate phase modulation and improved diffraction efficiency.
- Addressed membrane deformation, process repeatability, and fabrication uniformity challenges.
- Supported scalable fabrication approaches for lightweight deployable optical systems.

Multilevel Photon Sieve Fabrication

- Designed and fabricated multilevel photon sieve structures for high-efficiency diffractive optics.
- Implemented grayscale lithography and nanofabrication techniques for subwavelength feature control.
- Optimized fabrication parameters to improve optical performance and structural fidelity.

GaAs–Graphene Photodetectors

- Fabricated hybrid graphene–GaAs photodetectors for mid-IR applications using III-V semiconductor substrates.
- Developed integrated process workflows combining graphene transfer with GaAs semiconductor processing.
- Investigated carrier transport and interface effects impacting photodetector performance.
- Improved device responsivity, and fabrication repeatability through process optimization.

Si/SiO₂–Graphene Photodetectors

- Designed and fabricated graphene photodetectors on Si/SiO₂ platforms.
- Performed lithography, dielectric integration, contact engineering, and electrical characterization.
- Correlated fabrication parameters with optical and electrical device response.

Biosensor Development

- Developed nanostructured biosensors using surface-functionalized materials for enhanced sensitivity.
- Performed thin-film deposition and surface modification focused on interface-driven sensing.
- Applied data analysis and signal processing to improve detection accuracy and repeatability.
- Investigated material–biological interactions and long-term device stability.

Presentations & Workshops

- Poster Presentation: Fabrication of diffractive optical elements, international symposium on stochastic complex systems 2026
- Oral Presentation: Fabrication of Nanostructures on Polyimide Membranes for Diffractive Optical Elements, APS March Meeting 2026
- Oral Presentation: Fabrication of Nanostructures on Polyimide Membranes for Diffractive Optical Elements, Virginia Soft Matter Workshop X 2025.
- Oral Presentation: Fabrication of nanostructures for Diffractive optical elements, APS March Meeting 2024
- Poster Presentation: Diffractive optical elements on fused silica glass, Virginia Tech Soft Matter Symposium 2024

Collaborations

Graduate Research Collaboration with Dr. John Lecky, NASA Langley Research Center

PhD Advisor: Prof. Vinh Nguyen, Virginia Tech